Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

1750 – 2024 | Data from Global Carbon Budget 2025 & NASA

For policy analysts and climate negotiators

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Mauna Loa, February 2026

~280 ppm

Pre-industrial baseline

>50%

Increase since pre-industrial era

- Current CO_2 is at least 1.5× greater than the natural increase observed at the end of the last ice age 20,000 years ago.
- That ice-age change took thousands of years — the current rise has occurred in roughly two centuries.
- The acceleration is linked primarily to burning coal, oil, and natural gas, confirmed by ground-based and satellite measurements.

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Bt to 35+ Bt

~6 Bt

1950 annual emissions

~20 Bt

1990 annual emissions

>35 Bt

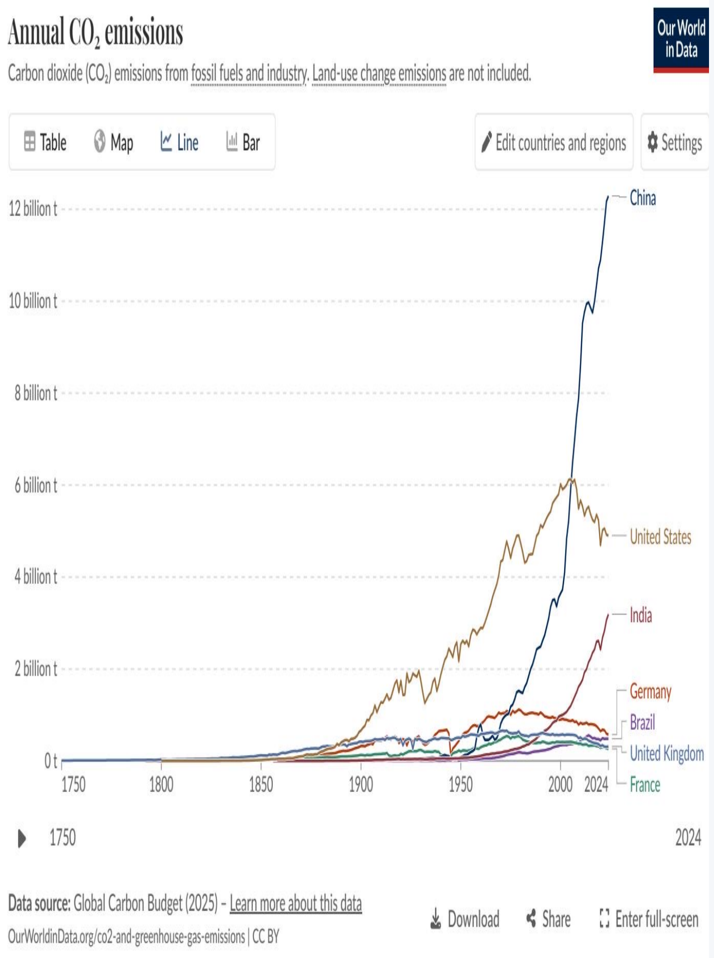
2024 annual emissions

- Before the Industrial Revolution, CO_2 emissions were negligible.
- Growth remained slow until the mid-20th century, then accelerated sharply.
- Emissions nearly quadrupled from 1950 to 1990, then almost doubled again by 2024.
- While growth rates have slowed in recent years, global emissions have not yet peaked.

See figure_3.jpg for the annual CO_2 emissions by country time series.

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Country / Region	Annual CO_2	Global Share	Recent Trend
China	~12.5 Bt	~27%	Slight increase
United States	~5.0 Bt	~14%	Declining
India	~3.0 Bt	~8%	Rising
EU-27	~2.7 Bt	~7%	Declining
Germany	~0.6 Bt	~1.7%	Declining
United Kingdom	~0.3 Bt	~0.9%	Declining
Brazil	~0.5 Bt	~1.4%	Stable
France	~0.3 Bt	~0.8%	Declining



Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900

Europe + US
>90%

Asia
<5%

Rest of World
~5%

2024

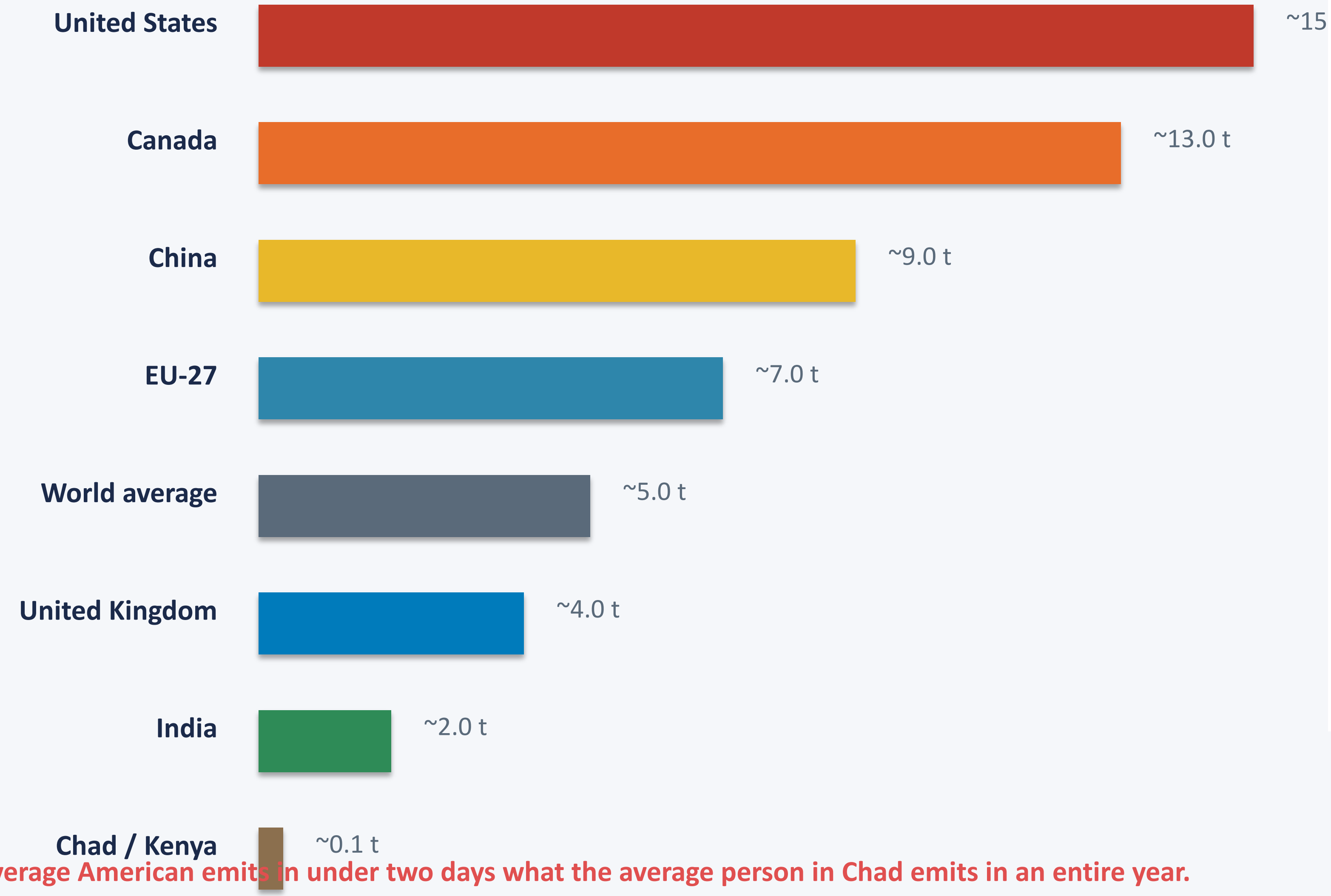
Asia
~50%

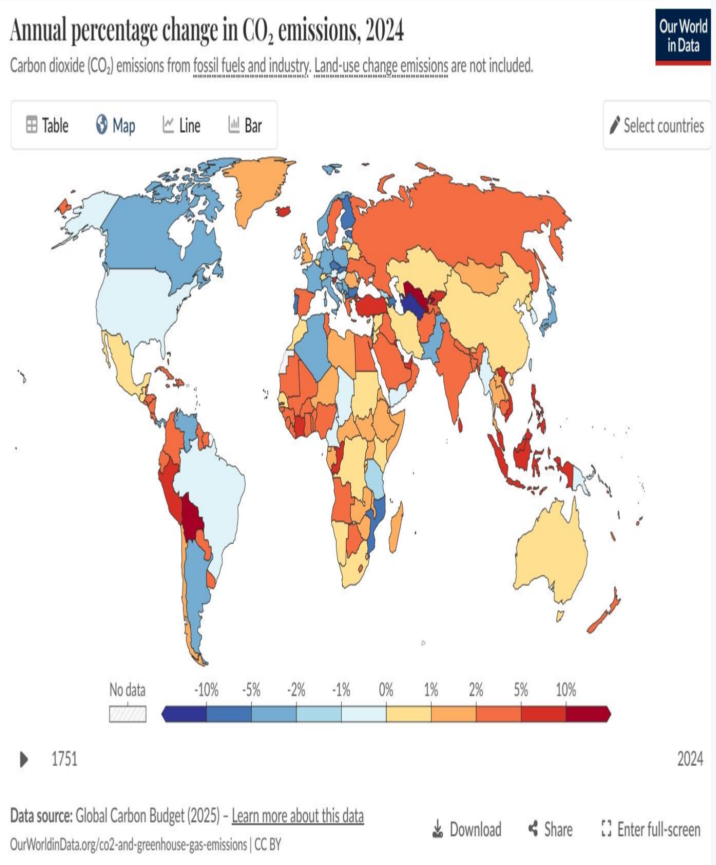
Europe + US
<33%

Africa & S. America
~3–4% each

Africa and South America each contribute only 3–4% of global emissions — comparable to international aviation and shipping combined. In 1950, Europe and the US still represented over 85% of global emissions.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

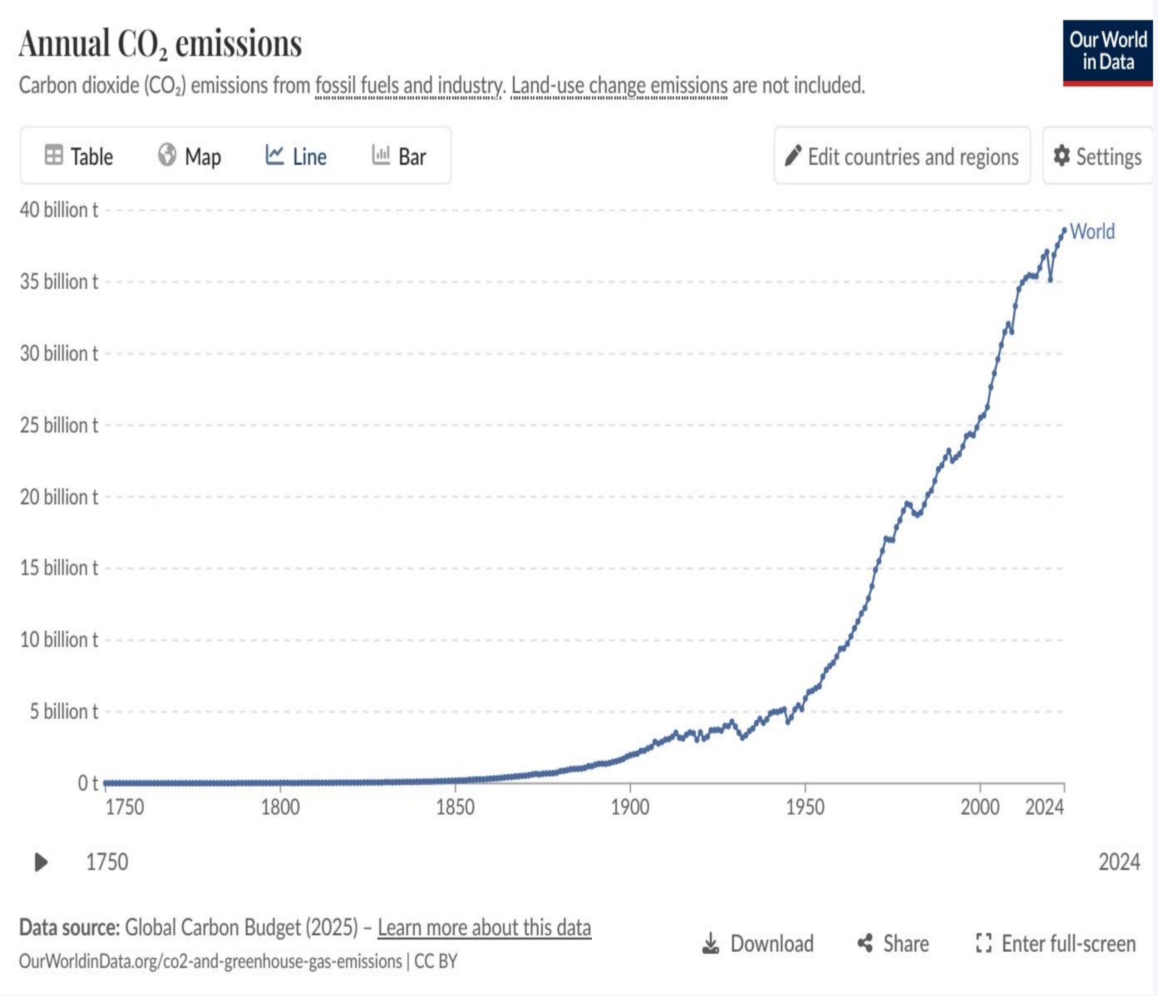




The average American emits in under two days what the average person in Chad emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

- United States and much of Europe: declines of −1% to −5% (shown in blue on the map).
- Russia, parts of Southeast Asia, and several African nations: increases of +2% to +10% (orange/red).
- Some South American countries experienced very large percentage increases (>10%).
- The mixed picture shows that while some major economies are decarbonizing, global emissions have not yet peaked.



Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

- Countries with similar living standards can have very different per capita emissions depending on their energy sources.
- Nuclear and renewable energy dramatically reduce per capita CO_2, even at high levels of prosperity.

Country	Key Energy Sources	Per Capita CO_2
France	High nuclear (~70%), growing renewables	~4 t/year
United Kingdom	Nuclear + offshore wind, declining coal	~4 t/year
Germany	Higher fossil fuel share, coal phase-out ongoing	~8 t/year*
Netherlands	Natural gas dependent, offshore wind growing	~8 t/year*

* Approximate; Germany's per capita figure reflects its overall energy mix including coal and industrial base.

Key Takeaways

1. Global CO_2 emissions now exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
2. China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is rising fast.
3. The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today.
4. Per capita inequalities are extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t) from the lowest (Chad, Niger at ~0.1 t).
5. Energy mix and policy choices matter: countries with similar prosperity (e.g., France vs. Germany) can have very different per capita emissions.